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RESEARCH ARTICLE

STRCFD: Small Maneuvering Object Tracking via Improved STRCF and Redetection in Near Infrared Videos

KUN QIAN^{ID}, (Member, IEEE), JIU-SHAN WANG, AND SHOU-JIN ZHANG

School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, Jiangsu 214122, China

Corresponding author: Kun Qian (kqian@jiangnan.edu.cn)

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ABSTRACT Tracking small objects in infrared videos is challenging due to a complex background, weak information and target mobility. To deal with these difficulties, an infrared small target tracking algorithm is proposed, which utilizes the Spatial-Temporal Regularized Correlation Filter (STRCF) as the backbone. First, the local image patch that refers to the target and its neighborhood background is given in the STRCF. Then, the guided local contrast mechanism is designed to eliminate noise and distinguish the target from the background. Furthermore, the robustness of the tracking is improved by using the detection model with an adaptive factor of search range, helping to alleviate the problem of tracking migration. Experimental results on entire public near-infrared videos show the superior performance of the proposed algorithm (named STRCFD), compared to several related algorithms in visual effects and objective evaluation. It should be mentioned that the proposed STRCFD achieves an overall precision of 81.1%.

INDEX TERMS Small object detection and tracking, near-infrared videos, correlation filter, guided filtering, local contrast.

I. INTRODUCTION

Infrared small target tracking holds significant importance in various fields, particularly in military, security, industrial, medical, and other domains. For effective tracking of small infrared targets, continuously estimating the state of the target is difficult due to the complex environment, the fast motion, and false targets with a similar representation [1], [2], [3]. Furthermore, the mechanism to improve the efficiency of the tracking algorithm should be carefully considered. Therefore, the small maneuver target tracking algorithm still needs further development [2], [4], [5], [6].

Traditional small target tracking methods contain mainly Detect Before the Track (DBT) [7], [8] and Track Before Detect (TBD) [9], [10], [11], [12]. The DBT method emphasizes the importance of detecting and identifying potential threats or targets as early as possible before starting

the tracking. It utilizes spatio-temporal filtering to eliminate discontinuous clutter. However, this method cannot obtain the information about the target because of the false target. The TBD method focuses on initiating the tracking of an object before detecting it. This approach is often used in situations where the target is already known, and tracking is used to continuously monitor its movement. In practical terms, the DBT method may be used in scenarios such as tracking the movement of known aircraft. The emphasis is on using the accumulation of energy in the target trajectory to improve the signal-to-noise ratio [11]. However, it is time-consuming to perform the comprehensive search for targets. In summary, two types of algorithms face detection and recognition problems [13], [14], due to strong background clutter and dim targets.

Subsequently, the object tracking methods focus on updating the model online. Early online tracking algorithms are derived from Discriminative Correlation Filter (DCF) methods [2], [15], [16], [17], [18], [19], [20], [21], [22],

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which have achieved great success. The DCF method obtains a correlation filter that identifies the object and its neighborhood with a fast Fourier transform [23]. The improved algorithms contain the Kernelized Correlation Filter (KCF) [16], and discriminative scale space tracking [24], which take advantage of the HOG feature. However, these algorithms may have the problem of overfitting. To solve this problem, DCF algorithms are improved by adding a constraint to the loss function, such as the Spatial Regularized Discriminative Correlation Filter (SRDCF) [17] and the Spatial Temporal Regularized Correlation Filter (STRCF) [18], [19]. However, the aforementioned DCF methods may fail when referring to small targets with little information.

To improve the performance of the small target tracking algorithm, background suppression and target detection have been the main mechanisms in recent years. The background suppression mechanism tends to improve image quality before feature extraction. Qian et al. [25] design a background suppression method that takes advantage of the Singular Value Decomposition (SVD). However, this algorithm is sensitive to some complex background, because the parameter is set manually. On this basis, the Guided Image Filter (GIF) is used to obtain the enhanced feature of a small target [25]. However, it should be noted that the guide filtering mechanism may not be sensitive to noise or edges that share a similar appearance to the target. There are other background suppression methods, including morphological filtering [26] and Sparse Representation (SR) [27]. Yet, the morphological method depends on the template, and the real-time performance of the SR methods is inefficient. Moreover, Huo et al. [28] propose a multi-feature fusion model, which is embedded in the particle filter for small target tracking. Once the target moves outside the search region, the detection mechanism [29], [30], [31] can be effective in tracking migration. Qian et al. first attempted to incorporate detectors in small target tracking. Nevertheless, the Kernelized Correlation Filter and Detection (KCFD) method [32] by pipeline filtering is time-consuming. Here, the feature enhancement method and the target redetection method encourage us to improve the performance of small target tracking.

Based on the analysis presented above, this paper proposes a STRCF based algorithm to track small targets in Near-Infrared (NIR) videos. Specifically, to enhance target information, a cascade filter based on GIF is designed to suppress strong edges and stochastic noise. Besides, an adaptive factor that refers to the search range is designed to help targets move quickly. In addition, an effective detection mechanism is designed to find the lost target again. More details are available in Section II and Section III. Experimental results show that the proposed algorithm can accurately track small infrared targets in the sky or the ground background. The main contributions are summarized below.

- The small target tracking is proposed for the Unmanned Aerial Vehicle (UAV) on NIR videos.

- A target enhancement method is designed based on the Guided Local Contrast (GLC), which helps to obtain good responses for small targets.
- An adaptive factor that refers the search range is designed for targets' moving fast. In addition, a tracking model with high robustness is designed utilizing a re-detection mechanism.

The main contents of this paper are as follows. Section I gives the introduction. Section II describes the related work that refers to the proposed algorithm. Section III contains the details of the tracking method and migration analysis. Section IV gives the experimental result and analysis, and the conclusion is drawn in Section V.

II. RELATED WORK

This section will review three main aspects relevant to our work, the DCF tracking methods, the enhancement methods, and the redetection mechanism.

A. THE DCF METHODS

Recently, DCF based methods [2], [15], [16], [17], [18], [19], [20], [21], [22] perform well in the field of object tracking. To be specific, the minimum output sum of the squared error algorithm [15] uses the gray feature but cannot follow the target with a large change in size. The KCF [16] extracts the HOG features and obtains a high speed with the kernel trick. However, the boundary effect must be suppressed, which can be solved by the SRDCF [17]. Furthermore, the STRCF [18], [19] adds time-regular terms based on the SRDCF, preventing rapid corruption of the model. On the basis of this, we introduce the STRCF based algorithm for tracking small UAV targets. The main difficulty of this task lies in the effective enhancement of target information and the stability of the tracking model, which are described in the next two sections.

Furthermore, Deep Learning (DL) [33], [34], [35], [36] technology is widely used in the field of object tracking. The popular transformer [33], [36] and the siamese network [37] are effective due to training with big data. Yet, there are almost no small target tracking methods based on DL models. We designed a Siamese-based algorithm for Infrared Small target Tracking (SiamIST) [38]. The overall performance on the public data set is good, but fails to detect the target that moves very fast.

B. THE ENHANCEMENT METHODS

For infrared small target tracking methods based on the DCF model, the small target is too dim to be detected, especially when the background is complicated. Thus, it is necessary to enhance the information of the target, which can be achieved by suppressing the background, i.e. image filtering. The fast SVD method [25] is sensitive to different types of background. To solve this problem, a salience enhancement method inspired by the human visual mechanism [39] is adopted to extract local features. However, it does not perform well on strong infrared edges. To solve this problem, the

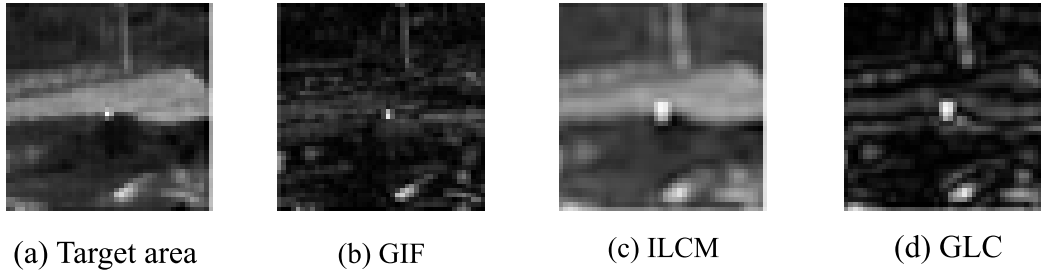


FIGURE 1. The effect of GLC over strong edge.

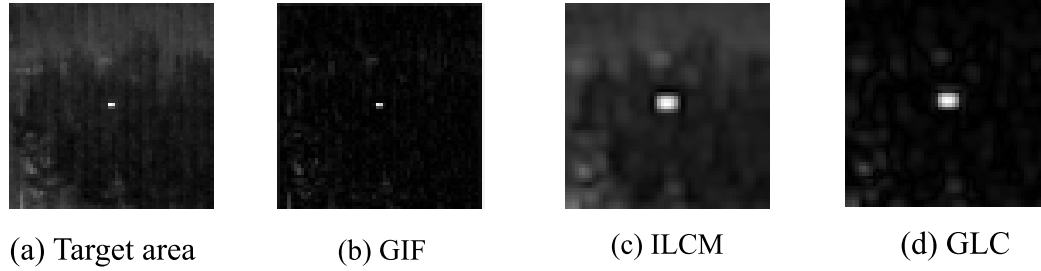


FIGURE 2. The effect of GLC referring to a dim-small target.

GIF [25] and the side window filter method [40] are proposed based on local filtering. The guide filtering mechanism works well thanks to the guide image. However, it should be noted that it may not be sensitive to noise that shares a similar appearance to the target. Therefore, in this article, we design a target enhancement model, namely the guided local contrast method, which combines the GIF and the local salience enhancement model. This model can increase the local contrast of the target image and suppress the interference of the edge in the target tracking.

C. THE DETECTION MECHANISM

To deal with migration tracking, an effective detector [29], [30], [31] is added to obtain the target outside the search region. Target detection methods are divided mainly into sequence-based methods [41], [42] and single frame detection methods [43], [44], [45]. Sequence-based methods analyze temporal patterns over a series of frames. For example, background modeling [41] methods and interframe difference [42] methods use spatial-temporal information to capture the target trajectory. However, the performance is sensitive to the situation where the target moves rapidly on a complex background. Moreover, it is difficult for the real-time performance of this detection algorithm to meet real-time requirements.

Single frame detection methods are designed to identify objects within a standalone image. Such algorithms are more rapid to aid in real-time tracking, including background suppression [43] based methods, DL [44] based methods, and SR based algorithms [45]. Here, background suppression is sensitive to strong background clutter in infrared videos, and

DL algorithms need a large number of labeled images for training. Therefore, the SR-based algorithm is added to the proposed tracking algorithm to find the target.

III. MATERIALS AND METHODS

A. THE STRCF METHOD

With two regular terms, the STRCF focuses on obtaining the optimal correlation filter model f_t^* , which can be expressed as

$$f_t^* = \arg \min_{f_t} \frac{1}{2} \left\| \sum_{d=1}^D x_t^d * f_t^d - y \right\|^2 + \frac{1}{2} \sum_{d=1}^D \|w * f_t^d\|^2 + \frac{\mu}{2} \|f_t - f_{t-1}\|^2 \quad (1)$$

where x_t^d and y represent the d -th dimension of the sample at the t -th frame and its corresponding response. D is the maximum value of the characteristic dimension d . In addition, $\frac{1}{2} \sum_{d=1}^D \|w * f_t^d\|^2$ and $\frac{\mu}{2} \|f_t - f_{t-1}\|^2$ denote the spatial and time regular terms, respectively. w and μ are related coefficients.

With the alternating-direction method of multipliers, Equation 1 can be transformed into several sub-problems.

$$f_t^{(i+1)} = \arg \min_{f_t} \left\| \sum_{d=1}^D x_t^d * f_t^d - y \right\|^2 + \gamma \|f_t - g + h\|^2 + \mu \|f_t - f_{t-1}\|^2 \quad (2)$$

$$g^{(i+1)} = \arg \min_g \sum_{d=1}^D \|w * g^d\|^2 + \gamma \|f_t - g + h\|^2 \quad (3)$$

$$h^{(i+1)} = h^i + f_t^{(i+1)} - g^{(i+1)} \quad (4)$$

where γ is a step parameter, and $h = s/\gamma$ (s is Lagrange multiplier). $g = f_t$ is an auxiliary variable constructed.

In the Fourier domain, the optimal correlation filter model $\hat{f}_t^*$ can be represented as

$$\hat{f}_t^* = \arg \min_{\hat{f}_t} \left\| \sum_{d=1}^D \hat{x}_t^d \cdot \hat{f}_t^d - \hat{y} \right\|^2 + \gamma \left\| \hat{f}_t - \hat{g} + \hat{h} \right\|^2 + \mu \left\| \hat{f}_t - \hat{f}_{t-1} \right\|^2 \quad (5)$$

where $\hat{\cdot}$ denotes the frequency symbol. Furthermore, the j -th element of the label $\hat{y}$ is only based on the j -th element of the sample $\hat{x}_t$, and the filter $\hat{f}$ across all D channels, which is represented by $V_j(\hat{f})$. Then, Equation 5 can be rewrite as

$$V_j(\hat{f}_t^*) = \arg \min_{V_j(\hat{f}_t)} \left\| V_j(\hat{x}_t)^T V_j(\hat{f}_t) - \hat{y}_j \right\|^2 + \gamma \left\| V_j(\hat{f}_t) - V_j(\hat{g}) + V_j(\hat{h}) \right\|^2 + \mu \left\| V_j(\hat{f}_t) - V_j(\hat{f}_{t-1}) \right\|^2 \quad (6)$$

The closed-form solution for $V_j(\hat{f}_t^*)$ is calculated with the derivative of Equation 6 being zero.

$$V_j(\hat{f}_t^*) = \frac{1}{\mu + \gamma} \left(I - \frac{V_j(\hat{x}_t) V_j(\hat{x}_t)^T}{\mu + \gamma + V_j(\hat{x}_t)^T V_j(\hat{x}_t)} \right) q \quad (7)$$

where $q = V_j(\hat{x}_t) \hat{y}_j + \gamma V_j(\hat{g}) - \gamma V_j(\hat{h}) + \mu V_j(\hat{f}_{t-1})$.

With the Fourier transform, the computational complexity of Equation 7 is much lower than Equation 1, which helps to increase the efficiency of the proposed algorithm.

B. THE MIGRATION ANALYSIS

For the infrared small target tracking, there are several difficulties containing small size, background clutter, and fast motion [46]. In low-resolution images, the target lacks texture features, and the frequency information of the background edge or noise is similar to that of a small target. In this situation, the tracking model may mistake the noise as the target, and then the background information is incorporated into the matching template. Furthermore, the camera shake problem often causes the phenomenon of wide-range movement of the target. When the target deviates from the previous search region, it is difficult for DCF algorithms to continuously detect the target.

Thus, it is necessary to find an effective method which should enhance the information of small targets and simultaneously reduce the influence caused by background clutter. In addition, online updating should be stopped while the tracker begins to lose the object of interest.

C. STRCFD: SMALL OBJECT TRACKING ALGORITHM BASED ON IMPROVED STRCF AND REDETECTION MODEL

1) TARGET ENHANCEMENT BASED ON GUIDED LOCAL CONTRAST

It is highly important to enhance the target's feature. Both the background suppression approach and the SR approach may not effectively differentiate targets from background clutter. Furthermore, a salience enhancement method inspired by the

human visual mechanism [39] is adopted to extract local features. However, it does not perform well on strong infrared edges. Therefore, in this article, a guided local contrast method is designed. First, the Improved Local Contrast Measure (ILCM) [47] aims to improve the salience of the target region. Then, the original information referring to the search region is set as the guide image, processing the salience map referring to ILCM.

In ILCM, a local image patch is divided into nine areas of 3×3 , and the local contrast ratio c_i^n can be expressed as

$$c_i^n = \min_i((L_n)^2/m_i) \quad (8)$$

where m_i denotes the gray mean of the i -th cell, L_n is the maximum of the gray value of the central cell m_0 in the n -th image patch.

Subsequently, the guided filter is adopted to suppress the influence of the background edge. With a local linear model, the Guided Image Filter (GIF) algorithm [48] generates an output considering the content of a guide image. Given an input image p_i and a guidance image G_i , the output image q_i is a linear transform of G_i in a window w_k , which is centered on the pixel k .

$$q_i = a_k G_i + b_k, \forall i \in w_k \quad (9)$$

where a_k and b_k are linear coefficients assumed to be constant in a box w_k with the radius being r . This local linear model ensures that q has an edge only if G has an edge, which is due to $\nabla q = \nabla G$, where ∇ represents the image gradient.

To determine linear coefficients, a cost function E that minimizes the difference between p and the filtered result q is represented by

$$E(a_k, b_k) = \sum_{i \in w_k} ((a_k G_i + b_k - p_i)^2 + \varepsilon a_k^2) \quad (10)$$

where ε is a regularization parameter, and $\sum$ denotes the sum operation. The solution can be obtained by minimizing Equation 10 with linear regression.

As shown in Figure 1 and Figure 2, an experiment is carried out to assess the noise reduction and edge preservation capabilities of the designed GLC in NIR images. Radius r is set to 4.

Figure 1 illustrates that the GLC is capable of suppressing background information referring to strong background edges. In addition, the GLC effectively enhances the information about the small target. However, utilizing a guided filter alone for image processing may not yield satisfactory results. In Figure 2, the number of pixels in the plane in the target region is less than four, which is difficult to extract features from.

2) ADAPTIVE SEARCH RANGE

In the STRCF method, the size of the search region is several times (nm) of the target region. For small target tracking, nm is 4, which takes into account background clutter and the algorithm's efficiency. Yet, the target has high

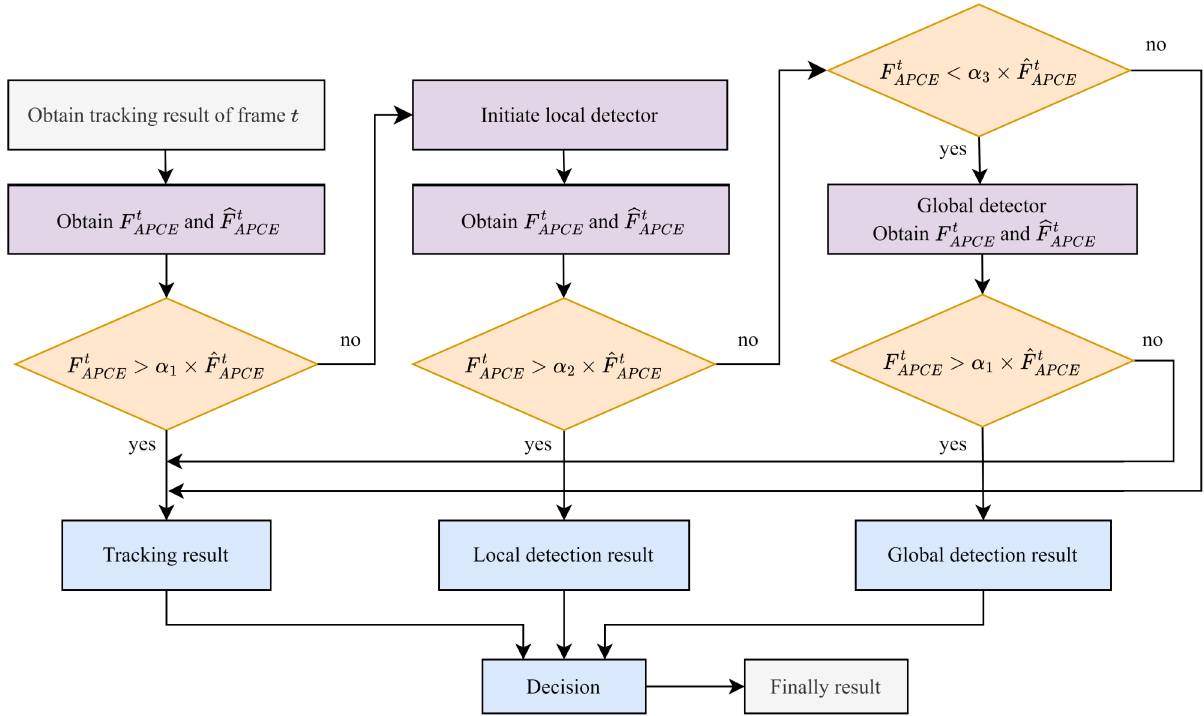


FIGURE 3. The detection mechanism.



(a) Frame 72



(b) Frame 73

FIGURE 4. The effect of the detection mechanism.

mobility and easily escapes the field of view. Therefore, it is necessary to expand nn once the target undergoes migration. Therefore, the proposed algorithm introduces the effective factor Average Peak-to-Correlation Energy (APCE), which denotes whether the target is in the tracking box.

$$F_{APCE} = \frac{|F_{max} - F_{min}|^2}{mean(\sum_x (F_x - F_{min})^2)} \quad (11)$$

where F_{max} , F_{min} and F_x denote the maximum value, the minimum value and the location x of the image gray, respectively. The degree of oscillation F_{APCE} of an image

suddenly decreases with the tracker losing targets in a short time.

With uncertainty of the tracking scenarios and difficulty, F_{APCE} fluctuates at different values. Therefore, an average APCE is designed in this paper to obtain a stable model, which is defined as

$$\hat{F}_{APCE}^t = \frac{\sum_{J=2}^t F_{APCE}^J}{t-1} \quad (12)$$

where F_{APCE}^J represents the APCE value at J -th frame. To be specific, $F_{APCE}^t < \alpha \hat{F}_{APCE}^t$ means that the tracking result referring to frame t is not reliable. Then, the parameter nn is

set as a larger number (such as 6), to obtain the information of the target.

3) THE DETECTION MODEL

Due to the fact of tracking migration, an effective detector [29], [30], [31], named the Partial Sum of the Tensor Nuclear Norm (PSTNN) [49], is utilized to detect the small target outside the search region. This algorithm focuses on the Image-Patch Tensor (IPT) model [50], the PSTNN model, and a weight that describes the local structure prior to the target. Specifically, the IPT model is a framework to separate the target from the background.

$$\mathcal{I} = \mathcal{B} + \mathcal{T} + \mathcal{N} \quad (13)$$

where $\mathcal{I}$ denotes the original image patch tensor, $\mathcal{B}$ denotes the background patch tensor with low rank. $\mathcal{T}$ denotes the sparse target patch tensor and $\mathcal{N}$ represents the noise patch tensor.

In fact, the noise patch tensor $\mathcal{N}$ is weak relative to $\mathcal{T}$, the problem of infrared small target detection can be transformed into the problem of separating low-rank and sparse tensors.

$$\min_{\mathcal{B}, \mathcal{T}} \text{rank}(\mathcal{B}) + \lambda \|\mathcal{T}\|_0 \quad s.t. \quad \mathcal{I} = \mathcal{B} + \mathcal{T} \quad (14)$$

where $\text{rank}(\cdot)$ represents rank of a matrix, and $\|\cdot\|_0$ is the nonzero entry in the matrix. Also, λ denotes a regular term.

Furthermore, with the PSTNN model and a weight related to the target prior, Equation 14 can be expressed as

$$\min_{\mathcal{B}, \mathcal{T}} \|\mathcal{B}\|_{PSTNN} + \lambda \|\mathcal{T} \odot \mathcal{W}\|_1 \quad s.t. \quad \mathcal{I} = \mathcal{B} + \mathcal{T} \quad (15)$$

where $\|\mathcal{B}\|_{PSTNN} = \sum_{i=N+1}^{\min(m,n)} \sigma_i(X)$ denotes partial sum of singular values of $\mathcal{B}$, which represents the low rank information of background. In addition, $\|\cdot\|_1$ is an approximate substitution for $\|\cdot\|_0$. $\mathcal{W}$ is a weight obtained from the local geometric features of the image, which describes the prior information of the target.

Once the tracker loses the target, the detection model is activated based on the APCE. In theory, it is considered that the tracking result are not trustworthy while the APCE of t -th frame satisfies $F_{APCE}^t < \alpha \times \hat{F}_{APCE}^t$. Furthermore, a coarse to fine strategy is designed to effectively deal with the situation of losing the target, which is shown in Figure 3.

To be specific, the detection mechanism is conducted with α_1, α_2 and α_3 , where $\alpha_3 < \alpha_1 < 1 < \alpha_2$. Besides, the state of a target can be obtained with the following steps.

- (1) The confidence of the tracking result is evaluated through comparing the F_{APCE}^t to the threshold $\alpha_1 \times \hat{F}_{APCE}^t$. The tracking result is unreliable while F_{APCE}^t is no greater than $\alpha_1 \times \hat{F}_{APCE}^t$, and local detection model is initiated.
- (2) With local detection, the target has a high probability of being detected. Moreover, $F_{APCE}^t > \alpha_2 \times \hat{F}_{APCE}^t$ is utilized to verify the validity of the detection method. Otherwise, the target should be detected in the global scope of images according to the parameter α_3 .

- (3) Global detection is enabled with the condition of $F_{APCE}^t < \alpha_3 \times \hat{F}_{APCE}^t$ is satisfied. Besides, the tracking result is updated by the detection result referring to $F_{APCE}^t > \alpha_1 \times \hat{F}_{APCE}^t$.

Subsequently, Figure 4 shows the effect of the detection mechanism.

The tracking box severely deviated from its true position at frame #72. At this point, the APCE value of the target region is lower than the corresponding threshold, activating the detector, and the target location is obtained at frame #73. Therefore, the tracking result is corrected.

4) THE DETAIL OF THE PROPOSED ALGORITHM

In this article, a STRCF based tracking framework is combined with a well-designed filter, an adaptive factor, and a detection model. Furthermore, the specific step of the proposed algorithm based on the improved STRCF (named STRCFD) is described below.

- (1) The target location is initialized and the search region $p_1(x)$ of the 1 th frame is obtained.
- (2) The GLC of $p_t(x)$ is obtained in each frame.

$$GLC(p_t(x)) = ILCM(p_t(x)) - GIF(ILCM(p_t(x)), p_t(x)) \quad (16)$$

where $p_t(x)$ denotes the search region at t -th frame, $ILCM$, GIF and GLC represent the local contrast filtering, the guided filter and guided local contrast filtering, respectively.

- (3) The response map s_t is computed using Equation 5

$$s_t = \text{corr}(GLC(p_t(x)), f_t) \quad (17)$$

where corr denotes the correlation function, and f_t is the filter at the t -th frame.

- (4) The APCE of STRCF is less than a fixed threshold, the search region is enlarged to some extent. (Here, the method without detection is named by STRCF2)
- (5) A coarse to fine detection mechanism is combined with the PSTNN detection model, and then the tracking result is corrected.

In short, the flow diagram of the proposed small target tracking algorithm in NIR videos is shown in Figure 5.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The experimental setup and data set are detailed in Section IV-A. Then, qualitative and quantitative comparisons with existing trackers in the entire NIR data set are given in Sections IV-B and IV-C, respectively. In addition, the ablation study is given in Section IV-D to prove the importance of GLC and detector in the proposed tracker.

A. EXPERIMENT SETUP

The NIR data set with wavelengths ranging from 3 μm to 5 μm , which was introduced by Hongqi Fan [51], can be found in <https://doi.org/10.11922/sciencedb.902>. A group of videos for a fixed-wing UAV target, which belongs to a

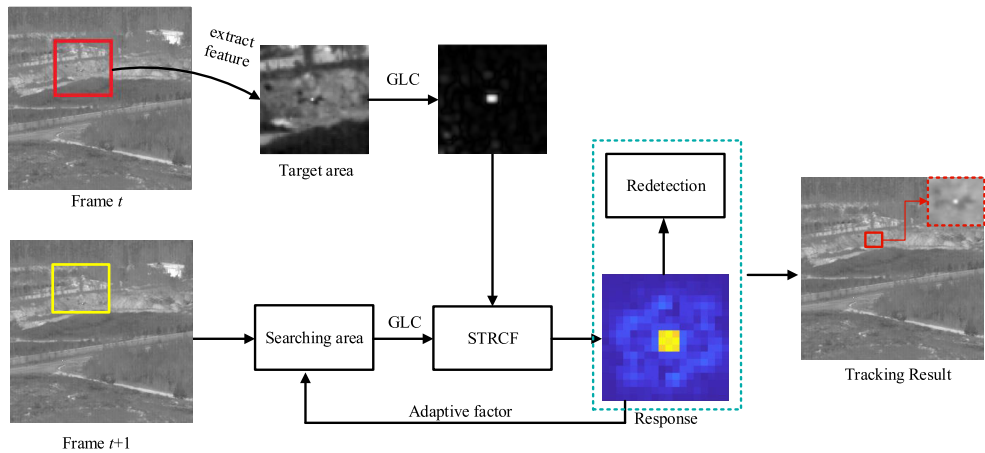
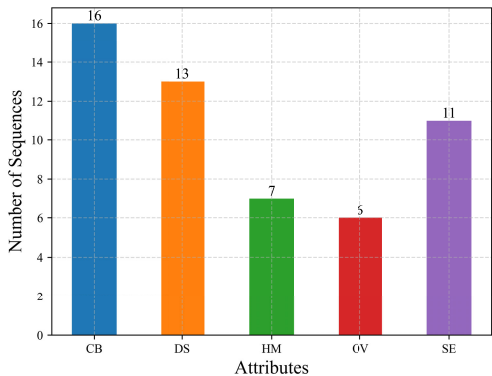


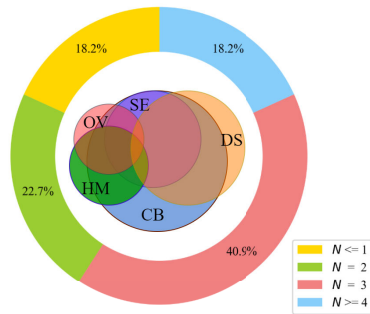
FIGURE 5. The flowchart of STRCFD.

TABLE 1. Description of six videos referring to Figures 7 to Figure 12 (quantitative comparison is achieved over the whole videos with the first 300 frames).

Videos	Average SCR	SCR Variance	Scene Description
data03	2.17	0.908	Close range, single target, open ground, out of view, reappearance.
data07	6.33	20.316	From near to far, single target, ground background.
data08	6.07	0.159	From far to near, single target, ground background.
data12	5.20	2.226	From far to near, single target, maneuvering midway, ground background.
data16	2.98	0.674	From far to near, single target, expanded targets, maneuvering, ground background.
data17	1.09	0.353	From near to far, single target, faint targets, ground background.



(a) The number of videos that refers to the attribute



(b) The distribution of all attributes

FIGURE 6. The distribution of the data set.

low-altitude flying target, is provided via outfield recording and post-data processing. The data acquisition scenario covers a sky background and a complex field background. In addition, the motion trajectories of the targets are unpredictable and change with different scenes. Figure 6(a) illustrates the number of videos that correspond to each attribute. Moreover, Figure 6(b) shows the distribution of various attributes, and the overlap between circles also indicates that a video corresponds to multiple attributes, whose number is represented by N . The data set can be encompassed by multiple attributes, such as Complex Background (CB), Dim

and Small (DS), High Mobility(HM), Out of View (OV), and Similar Edge (SE). In addition to these attributes, the single target occupies 93.8% of the total frames (16177), and the ground background accounts for 85.8%. Differences in shooting angles and faint targets also contribute to a sharp increase in tracking difficulty. As shown in Table 1, it describes the tracking scenarios for six sequences used in Section IV-B (SCR refers to the signal-to-cluster ratio). The data set is publicly released in series for the first time in the country, and these data scenarios include the following challenging issues in the detection and recognition of current

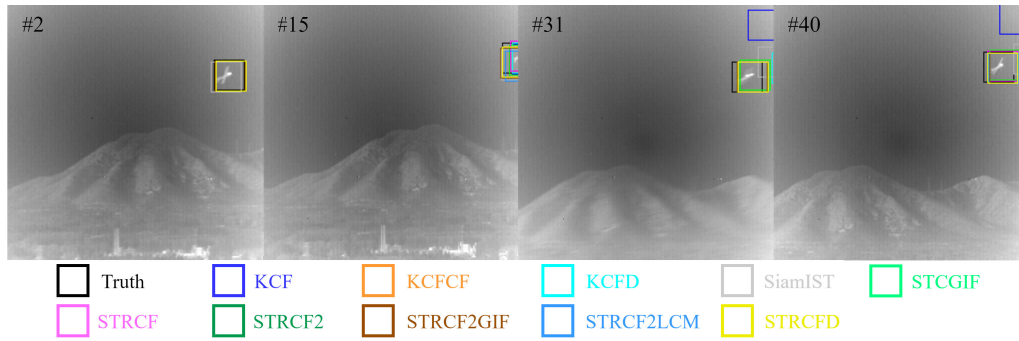


FIGURE 7. Qualitative results over *data03* video (Best viewed on color display).

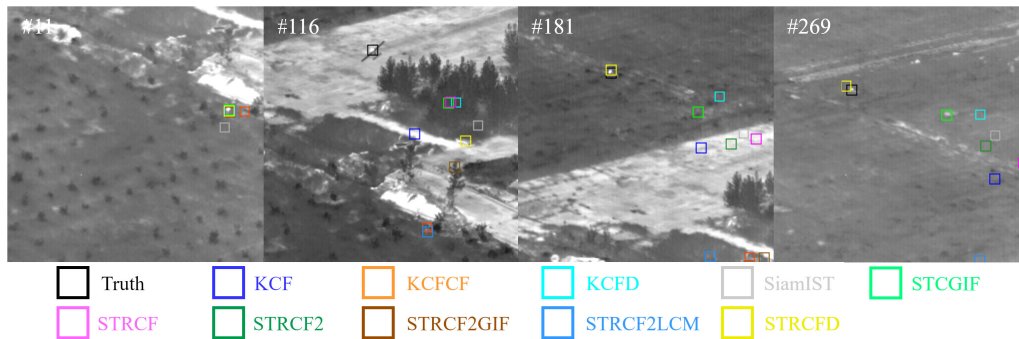


FIGURE 8. Qualitative results over *data07* video.

infrared targets, which needs to be addressed in future work.

- Target tracking in long sequence images.
- Detection and tracking under conditions of mutual interference among multiple targets.
- Discovery and capture of small targets in complex environments.

The publication of this data set effectively compensates for the lack of authenticity in infrared target simulation data and addresses the challenge of a shortage of measured data samples. Furthermore, this data set includes 22 image sequences, which can be used effectively to prove tracking performance.

To fully evaluate the performance of the proposed algorithms, some related IR trackers, including KCFD [32], SiamIST [38], STCGIF [52], KCF [16], KCFCF [25], and STRCF [18] are used to compare with STRCF2 (in section III-C4) and STRCFD. It should be noted that the results of the STRCF2GIF method and the STRCF2LCM method prove the performance of the GLC model, which is described in the ablation study section.

Generally, the size of the tracking box is set as 9×9 and the search region is set to 4 times that of the tracking box. In the GLC model, the radius and regularization parameters are set to 4 and 0.04, respectively. The spatial regular term coefficient w in Equation 1 ranges from 0.001 to 1000. In the detection model, $\alpha_1 = 0.5$, $\alpha_2 = 1.2$, and $\alpha_3 = 0.3$.

The simulation platform is Matlab2020a and the computer hardware is configured with a 2.1 GHz CPU and 16 Gb memory.

B. QUALITATIVE COMPARISON

In this section, the experimental results of all the test algorithms are shown in Figures 7 to Figure 11.

In the *data03* video, the small target is out of view, which is due to rotation of the IR device. Both KCF and KCFD lose the target at frame #31, which is unrecoverable. When SiamIST deals with large targets in this sequence, the parameter configuration is crucial for tracking accuracy.

In the *data07* video, the small target is submerged in a strong background clutter in the frame #116. In frame #11 of Figure 8, partial comparison algorithms accurately estimate the target location. However, all tracking algorithms are affected by ground background clutter, which is shown in the frame #116. Subsequently, only the STRCFD algorithm can detect the target in the frame #181 and the frame #269 continuously.

In the *data08* video, the target moves from far to near, and the background contains many edges that refer to the road. The KCF, STCGIF, and STRCF drift to the edge of the background after the frame #218, which is shown in Figure 9. The KCFCF always mislocates the target. Taking advantage of image filtering, the KCFD method and the proposed algorithm perform well in most images. And the

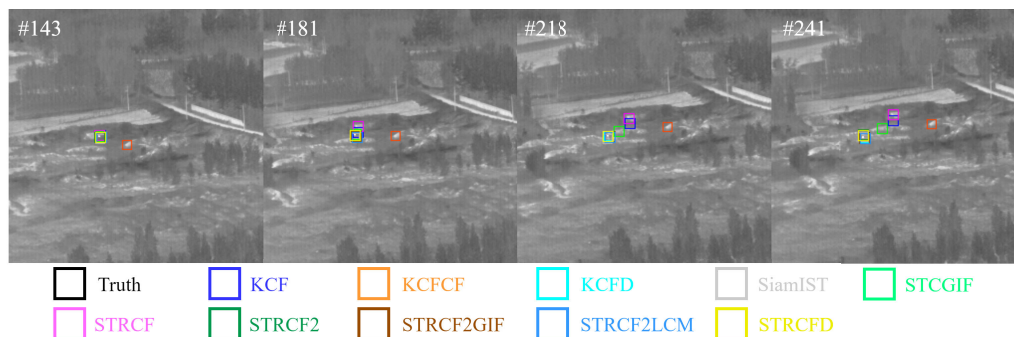


FIGURE 9. Qualitative results over *data08* video.

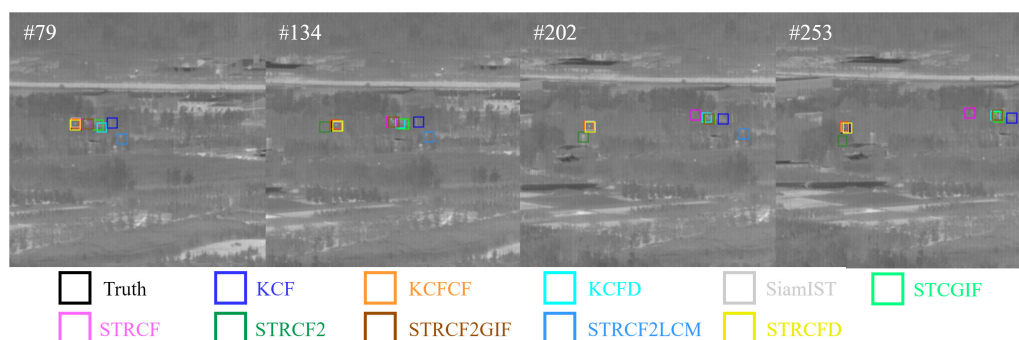


FIGURE 10. Qualitative results over *data12* video.

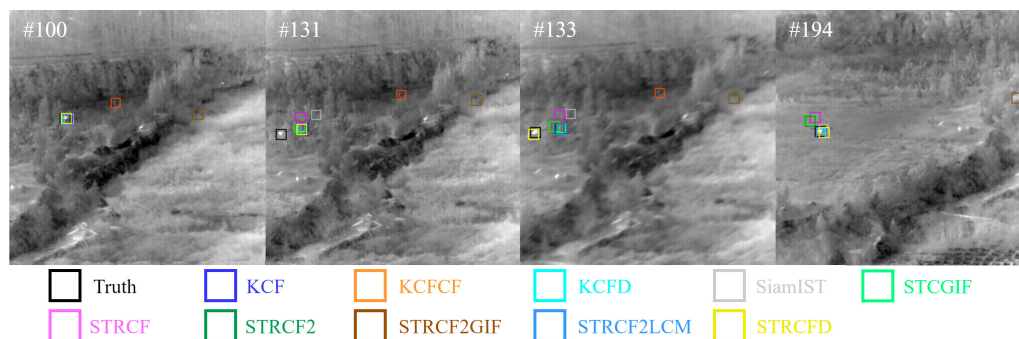


FIGURE 11. Qualitative results over *data16* video.

performance of SiamIST is stable in the bargain, because of data training.

In the *data12* video, the dim and small target moves fast at some point. As shown in Figure 10, the target undergoes a large displacement in a short time (around the frame #79). With the exception of some algorithms, most algorithms lose the target, which is outside the search region. In addition, the KCF algorithm is sensitive to noise in the background, which can be seen in Figure 10. It is worth noting that KCFCF and SiamIST can continuously track the target in this sequence. In summary, taking into account the APCE factor, the STRCFD algorithm obtains good tracking results on the *data12* images.

In the *data16* video, the background clutter is so strong. Specifically, most tracking algorithms lose the target in the frame #131 in Figure 11. However, the STRCFD algorithm can re-estimate the location of the target at the next frame, and continuously track the status of the target after the frame #133. Also, the same happens with *data03*, whose target is out of view (Figure 7).

In the video *data17*, the SCR referring to the small target is low, preventing accurate tracking. As shown in Figure 12, the dim target moves under the background that contains a strong edge. In frame #95, the SiamIST, the KCFCF, the KCFD and the STRCFD can overcome this difficulty. In addition, other algorithms do not track the target after the frame #95.

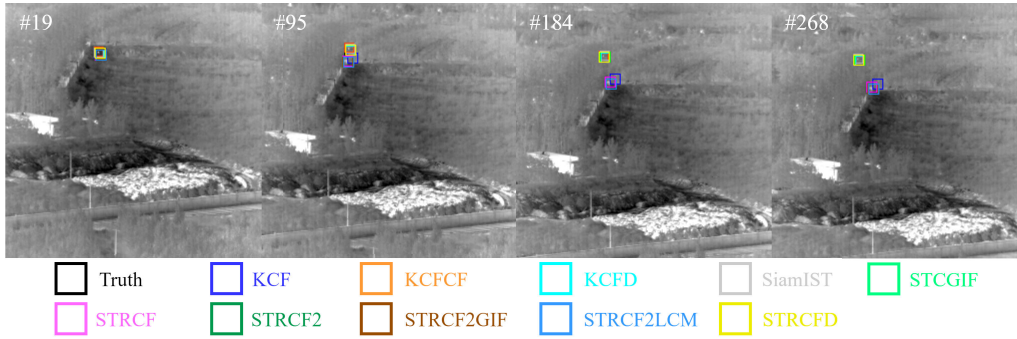


FIGURE 12. Qualitative results over *data17* video.

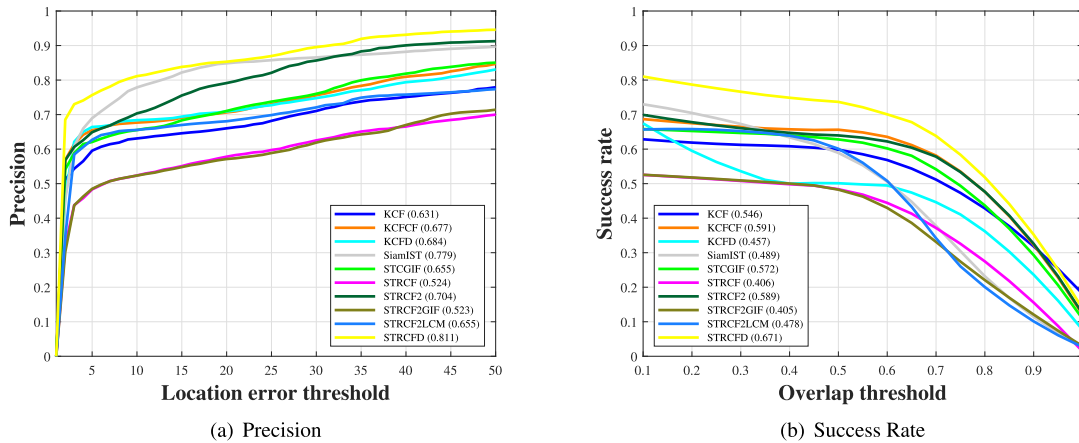


FIGURE 13. The performance curve.

Moreover, the STRCF algorithm mistakes the noise as a small target.

C. QUANTITATIVE COMPARISON

In this section, the precision plot and the success plot are used throughout the NIR data set [53] to accurately evaluate the performance of the proposed algorithms. Precision is treated as the ratio of successful frames whose tracking output is within the given threshold (in pixels) from the ground truth, measured by the center distance between bounding boxes. In addition, the success plot represents the proportion of successful frames, where the predicted tracking box overlaps the ground truth with a fixed threshold (from zero to one). Also, the curve is determined by the varying percentages that arise from different thresholds. The performance of all algorithms are quantified with the precision value referring to that the Center Location Error (CLE) d_{CLE} equals 10, and the Area Under the Curve (AUC) of the success curve.

With the initial target state, all the compared trackers are executed on all the test NIR videos. Subsequently, Figure 13 shows the performance curve of all NIR videos, and Table 2 shows the precision of six selected videos, referring to the threshold of 10 pixels. Furthermore, Table 3 shows the performance, which includes precision, success rate, and Frame Per Second (FPS), referring to all NIR videos.

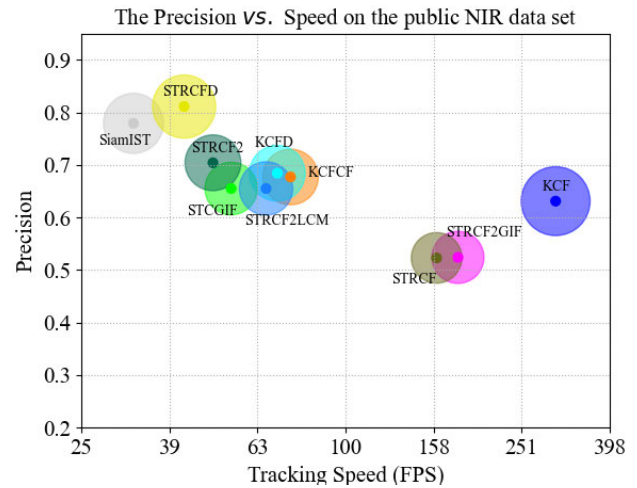


FIGURE 14. The Precision vs. Speed on the public NIR data set.

It can be observed from Figure 13 that the precision of the compared algorithms increases with the growth of the pixel spacing, whereas the success rate does not follow the same pattern entirely. The success rate tends to remain briefly at an approximate level at different overlap ratios. The DL-based SiamIST method exhibits satisfactory precision, but the

TABLE 2. The success rate (%) over the specific video ($d_{CLE} = 10$).

Videos	STCGIF	KCF	KCFCF	KCFD	SiamIST	STRCF	STRCF2	STRCFD
data03	79	25	27	50	79	75	77	84
data07	4	8	4	8	7	7	4	39
data08	69	64	96	77	70	20	96	97
data12	16	16	16	15	74	3	96	96
data16	45	92	67	60	65	78	84	88
data17	100	32	100	78	79	29	96	92
Average Precision	52	40	52	48	62	35	76	83

TABLE 3. Performance comparisons (%) over all NIR videos ($d_{CLE} = 10$).

Algorithm	AUC	Overall precision ($d_{CLE} = 10$)	Overall FPS
KCF	0.546	0.631	>300
KCFCF	0.591	0.677	75
KCFD	0.457	0.684	70
SiamIST	0.489	0.779	33
STCGIF	0.572	0.655	55
STRCF	0.406	0.524	180
STRCF2	0.589	0.704	50
STRCFD	0.671	0.811	43

TABLE 4. The ablation study.

Algorithm	AUC	Overall precision ($d_{CLE} = 10$)	Overall FPS
STRCF	0.406	0.524	180
STRCF2	0.589	0.704	50
STRCF2GIF	0.405	0.523	161
STRCF2LCM	0.478	0.655	66
STRCFD	0.671	0.811	43

success rate is not always perfect, particularly manifested in the instability of the tracking box. As shown in Figure 14, despite the passable performance in terms of precision and AUC, KCF stands out in terms of FPS due to its use of the fast Fourier transform.

As can be seen in Table 2, both the STRCF2 and the STRCFD obtain higher performance values except for the *data07* video (the precision value is low to 0.04), due to the fact that most trackers drift to strong background clutter. Furthermore, Table 3 shows that the STRCFD algorithm has a better tracking effect than other algorithms on all videos.

In summary, the tracking effect of the proposed algorithm is better both in subjective evaluation and in objective evaluation, which verifies that the proposed algorithm can better achieve tracking for small infrared targets.

D. ABLATION STUDY

To verify the effectiveness of the algorithm, an ablation experiment is performed using STRCF, STRCF2GIF, STRCF2LCM, STRCF2, and STRCFD. The results in the NIR data set are shown in Figures 7 to Figure 13. In Table 4, the AUC value of the baseline algorithm reaches 0.406. In addition, the algorithm with only the guided filter does not perform well on objective indicators. This is because it is difficult to enhance a dim target. On the contrary, with the guided local contrast measure, which effectively enhances the salience of the target, the AUC and precision

performance of STRCF2 are improved by 45% and 34%, respectively. Compared to the STRCF2 algorithm of the model without detection, the AUC value of STRCFD is improved by 14%, and the precision value ($d_{CLE} = 10$) increased by 15%. Due to the intervention of the redetection mechanism and other modules, the increase in computational load, coupled with the complexity added to the image processing, prevents FPS from reaching a state-of-the-art level.

In summary, the GLC model and the detection mechanism help to improve the accuracy and stability of target tracking on the public NIR dataset, which provides an effective method for the follow-up target tracking algorithm.

V. CONCLUSION

In this article, a tracking algorithm is proposed for small maneuvering targets in NIR videos. The GLC measure improves the target information for the tracking framework. In addition, the adaptive search factor helps to find the target continuously in a larger range. Most importantly, the experimental results insist that the detection model can improve tracking performance in the situation of losing targets. However, for STRCFD, the parameters in the redetection model must be established by experience to some extent. It should be mentioned that the proposed STRCFD achieves an overall precision of 81.1%. This lays the foundation for the development of future infrared small-target tracking algorithms.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

REFERENCES

- [1] M. Wan, X. Ye, X. Zhang, Y. Xu, G. Gu, and Q. Chen, "Infrared small target tracking via Gaussian curvature-based compressive convolution feature extraction," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [2] P. Yan, S. Yao, Q. Zhu, T. Zhang, and W. Cui, "Real-time detection and tracking of infrared small targets based on grid fast density peaks searching and improved KCF," *Infr. Phys. Technol.*, vol. 123, Jun. 2022, Art. no. 104181.
- [3] X. Dong, X. Huang, Y. Zheng, L. Shen, and S. Bai, "Infrared dim and small target detecting and tracking method inspired by human visual system," *Infr. Phys. Technol.*, vol. 62, pp. 100–109, Jan. 2014.
- [4] W. Li, L. Lv, and J. Zhu, "Multigroup spatial shift models for thermal infrared tracking," *Knowl.-Based Syst.*, vol. 255, Nov. 2022, Art. no. 109705.
- [5] D. Yuan, X. Shu, Q. Liu, and Z. He, "Structural target-aware model for thermal infrared tracking," *Neurocomputing*, vol. 491, pp. 44–56, Jun. 2022.
- [6] X. Li, S. Wang, W. Chen, Z. Weng, W. Fan, and Z. Tian, "An intelligent vision-based tracking method for underground human using infrared videos," *Symmetry*, vol. 14, no. 8, p. 1750, Aug. 2022.
- [7] F. Zhang, C. Li, and L. Shi, "Detecting and tracking dim moving point target in IR image sequence," *Infr. Phys. Technol.*, vol. 46, no. 4, pp. 323–328, Apr. 2005.
- [8] L. Deng, J. Zhang, G. Xu, and H. Zhu, "Infrared small target detection via adaptive M-estimator ring top-hat transformation," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107729.
- [9] M. Üney, P. Horridge, B. Mulgrew, and S. Maskell, "Coherent long-time integration and Bayesian detection with Bernoulli track-before-detect," *IEEE Signal Process. Lett.*, vol. 30, pp. 239–243, 2023.
- [10] W. Yi, Z. Fang, W. Li, R. Hoseinnezhad, and L. Kong, "Multi-frame track-before-detect algorithm for maneuvering target tracking," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 4104–4118, Apr. 2020.
- [11] A. Gnanasambandam and S. H. Chan, "Exposure-referred signal-to-noise ratio for digital image sensors," 2021, *arXiv:2112.05817*.
- [12] Z. Lu, P. Yueping, Y. Zecong, J. Rongqi, and Z. Tongtong, "Infrared small UAV target detection algorithm based on enhanced adaptive feature pyramid networks," *IEEE Access*, vol. 10, pp. 115988–115995, 2022.
- [13] R. Zhou and J. Teng, "An improved resampling algorithm for particle filtering in small target tracking," *J. Coastal Res.*, vol. 73, pp. 600–605, Mar. 2015.
- [14] X. Liu, L. Li, L. Liu, X. Su, and F. Chen, "Moving dim and small target detection in multiframe infrared sequence with low SCR based on temporal profile similarity," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [15] D. Bolme, J. R. Beveridge, B. A. Draper, and Y. M. Lui, "Visual object tracking using adaptive correlation filters," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 2544–2550.
- [16] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "High-speed tracking with kernelized correlation filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 583–596, Mar. 2015.
- [17] M. Danelljan, G. Häger, F. S. Khan, and M. Felsberg, "Learning spatially regularized correlation filters for visual tracking," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 4310–4318.
- [18] F. Li, C. Tian, W. Zuo, L. Zhang, and M.-H. Yang, "Learning spatial-temporal regularized correlation filters for visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4904–4913.
- [19] S. Feng, K. Hu, E. Fan, L. Zhao, and C. Wu, "Kalman filter for spatial-temporal regularized correlation filters," *IEEE Trans. Image Process.*, vol. 30, pp. 3263–3278, 2021.
- [20] N. Wang, W. Zhou, Q. Tian, R. Hong, M. Wang, and H. Li, "Multi-cue correlation filters for robust visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4844–4853.
- [21] S. Zhang, K. Zheng, and S. Huaiyuan, "Analysis of the occlusion interference problem in target tracking," *Math. Problems Eng.*, vol. 2022, pp. 1–10, Sep. 2022.
- [22] Q. Yu, K. Fan, and Y. Zheng, "Domain adaptive transformer tracking under occlusions," *IEEE Trans. Multimedia*, vol. 25, pp. 1452–1461, 2023.
- [23] Z. He, Z. Zhang, and C. Jung, "Fast Fourier transform networks for object tracking based on correlation filter," *IEEE Access*, vol. 6, pp. 6594–6601, 2018.
- [24] M. Danelljan, G. Häger, F. S. Khan, and M. Felsberg, "Discriminative scale space tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 8, pp. 1561–1575, Aug. 2017.
- [25] K. Qian, H. Zhou, S. Rong, B. Wang, and K. Cheng, "Infrared dim-small target tracking via singular value decomposition and improved kernelized correlation filter," *Infr. Phys. Technol.*, vol. 82, pp. 18–27, May 2017.
- [26] V. T. Tom, T. Peli, M. Leung, and J. E. Bondaryk, "Morphology-based algorithm for point target detection in infrared backgrounds," *Proc. SPIE*, vol. 1954, Oct. 1993, pp. 2–11.
- [27] T. Zhang, Z. Peng, H. Wu, Y. He, C. Li, and C. Yang, "Infrared small target detection via self-regularized weighted sparse model," *Neurocomputing*, vol. 420, pp. 124–148, Jan. 2021.
- [28] Y. Huo, Y. Chen, H. Zhang, H. Zhang, and H. Wang, "Dim and small target tracking using an improved particle filter based on adaptive feature fusion," *Electronics*, vol. 11, no. 15, p. 2457, Aug. 2022.
- [29] C. Liu, D. Wang, and C. Bo, "Re-detection and distractor association from a global perspective: A long-term tracking system," *Comput. Electr. Eng.*, vol. 107, Apr. 2023, Art. no. 108611.
- [30] J. Wang, H. Yang, N. Xu, C. Wu, Z. Zhao, J. Zhang, and D. O. Wu, "Long-term target tracking combined with re-detection," *EURASIP J. Adv. Signal Process.*, vol. 2021, no. 1, pp. 1–16, Dec. 2021.
- [31] N. Wang, W. Zhou, and H. Li, "Reliable re-detection for long-term tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 3, pp. 730–743, Mar. 2019.
- [32] K. Qian, S. H. Rong, and K. H. Cheng, "Anti-interference small target tracking from infrared dual waveband imagery," *Infr. Phys. Technol.*, vol. 118, Nov. 2021, Art. no. 103882.
- [33] F. Gu, J. Lu, and C. Cai, "RPformer: A robust parallel transformer for visual tracking in complex scenes," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–14, 2022.
- [34] D. Yuan, X. Chang, Q. Liu, Y. Yang, D. Wang, M. Shu, Z. He, and G. Shi, "Active learning for deep visual tracking," *IEEE Trans. Neural Netw. Learn. Syst.*, 2023.
- [35] F. Gu, J. Lu, C. Cai, Q. Zhu, and Z. Ju, "Repformer: A robust shared-encoder dual-pipeline transformer for visual tracking," *Neural Comput. Appl.*, vol. 35, no. 28, pp. 20581–20603, Oct. 2023.
- [36] F. Gu, J. Lu, and C. Cai, "A robust attention-enhanced network with transformer for visual tracking," *Multimedia Tools Appl.*, vol. 82, no. 26, pp. 40761–40782, Nov. 2023.
- [37] B. Li, J. Yan, W. Wu, Z. Zhu, and X. Hu, "High performance visual tracking with Siamese region proposal network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8971–8980.
- [38] K. Qian, S.-J. Zhang, H.-Y. Ma, and W.-J. Sun, "SiamIST: Infrared small target tracking based on an improved SiamRPN," *Infr. Phys. Technol.*, vol. 134, Nov. 2023, Art. no. 104920.
- [39] X. Zhang, Z. Xu, and H. Liao, "Human motion tracking and 3D motion track detection technology based on visual information features and machine learning," *Neural Comput. Appl.*, vol. 34, no. 15, pp. 12439–12451, Aug. 2022.
- [40] H. Yin, Y. Gong, and G. Qiu, "Side window filtering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8750–8758.
- [41] A. Zhou, W. Xie, and J. Pei, "Background modeling combined with multiple features in the Fourier domain for maritime infrared target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.
- [42] M. Wan, G. Gu, E. Cao, X. Hu, W. Qian, and K. Ren, "In-frame and inter-frame information based infrared moving small target detection under complex cloud backgrounds," *Infr. Phys. Technol.*, vol. 76, pp. 455–467, May 2016.
- [43] M. Zhao, L. Li, W. Li, R. Tao, L. Li, and W. Zhang, "Infrared small-target detection based on multiple morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6077–6091, Jul. 2021.
- [44] H. Zheng, J. Liu, and X. Ren, "Dim target detection method based on deep learning in complex traffic environment," *J. Grid Comput.*, vol. 20, no. 1, p. 8, Mar. 2022.
- [45] F. Yan, G. Xu, J. Wang, Q. Wu, and Z. Wang, "Infrared small target detection via Schatten capped pNorm-based non-convex tensor low-rank approximation," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
- [46] J. Yan, J. Du, Y. Young, C. R. Chatwin, R. C. D. Young, and P. Birch, "Real-time unmanned aerial vehicle tracking of fast moving small target on ground," *J. Electron. Imag.*, vol. 27, no. 5, Sep. 2018, Art. no. 053010.

- [47] Y. Dai, Y. Wu, F. Zhou, and K. Barnard, "Attentional local contrast networks for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9813–9824, Nov. 2021.
- [48] K. He, J. Sun, and X. Tang, "Guided image filtering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 6, pp. 1397–1409, Jun. 2013.
- [49] L. Zhang and Z. Peng, "Infrared small target detection based on partial sum of the tensor nuclear norm," *Remote Sens.*, vol. 11, no. 4, p. 382, Feb. 2019.
- [50] Y. Dai and Y. Wu, "Reweighted infrared patch-tensor model with both nonlocal and local priors for single-frame small target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3752–3767, Aug. 2017.
- [51] B. Hui, Z. Song, and H. Fan, "A dataset for infrared detection and tracking of dim-small aircraft targets under ground/air background," *China Sci. Data*, vol. 5, no. 3, pp. 291–302, 2020.
- [52] K. Qian, H. Zhou, and H. Qin, "Infrared dim-small target tracking based on guide filter and spatio-temporal context learning," *Acta Photonica Sinica*, vol. 44, no. 9, 2015, Art. no. 910003.
- [53] Q. Liu, Z. He, X. Li, and Y. Zheng, "PTB-TIR: A thermal infrared pedestrian tracking benchmark," *IEEE Trans. Multimedia*, vol. 22, no. 3, pp. 666–675, Mar. 2020.



JIU-SHAN WANG is currently pursuing the M.S. degree in software engineering with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China. His research interests include machine learning and IR target tracking.



KUN QIAN (Member, IEEE) received the B.S. and Ph.D. degrees in electronic science and technology from Xidian University, Xi'an, China, in 2012 and 2019, respectively. He visited Griffith University, Australia, for one year. He is currently a Lecturer with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China. His research interests include machine learning, computer vision, and hyperspectral/IR video tracking.



SHOU-JIN ZHANG received the B.S. degree in software engineering from Zhejiang Gongshang University, Hangzhou, China, in 2022. He is currently pursuing the M.S. degree in software engineering with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China. His research interests include deep learning and video tracking.

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